

B.Tech. Degree VII Semester Examination November 2013

EB/EC/EI 705 (D) MECHATRONICS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain the elements of Mechatronic system.
(b) Write short notes on optical encoders.
(c) Explain the working of a quick return mechanism.
(d) List the specifications of a stepper motor.
(e) Define the following.
 (i) Rise time
 (ii) Peak time
 (iii) Settling time
(f) Describe MEMS humidity microsensor briefly.
(g) Explain the role of intelligent controllers in mechatronics.
(h) Write short note on FMS.

PART B

(4 × 15 = 60)

- II. (a) What is Hall effect? Write short notes on Hall effect sensors. (5)
(b) Describe the factors to be considered in selecting a sensor. (10)
- OR**
- III. (a) Write short notes on temperature compensation. (5)
(b) Explain data acquisition system with a neat block diagram. (10)
- IV. (a) Write short notes on kinematic chains. (5)
(b) Explain the working of process control valves. (10)
- OR**
- V. Describe the various solid state devices that can be used electronically to switch circuits. (15)
- VI. (a) Explain the working of Rack-and-pinion. (5)
(b) Explain the fluid system building blocks. (10)
- OR**
- VII. Explain the fibre optic devices in mechatronics. (15)
- VIII. (a) Explain proportional plus derivative control with necessary circuits. (5)
(b) Explain the working of CNC machines. (10)
- OR**
- IX. (a) Explain the programmable logic controller with the help of a block diagram. (10)
(b) What is the role of robot in a mechatronic system? (5)
